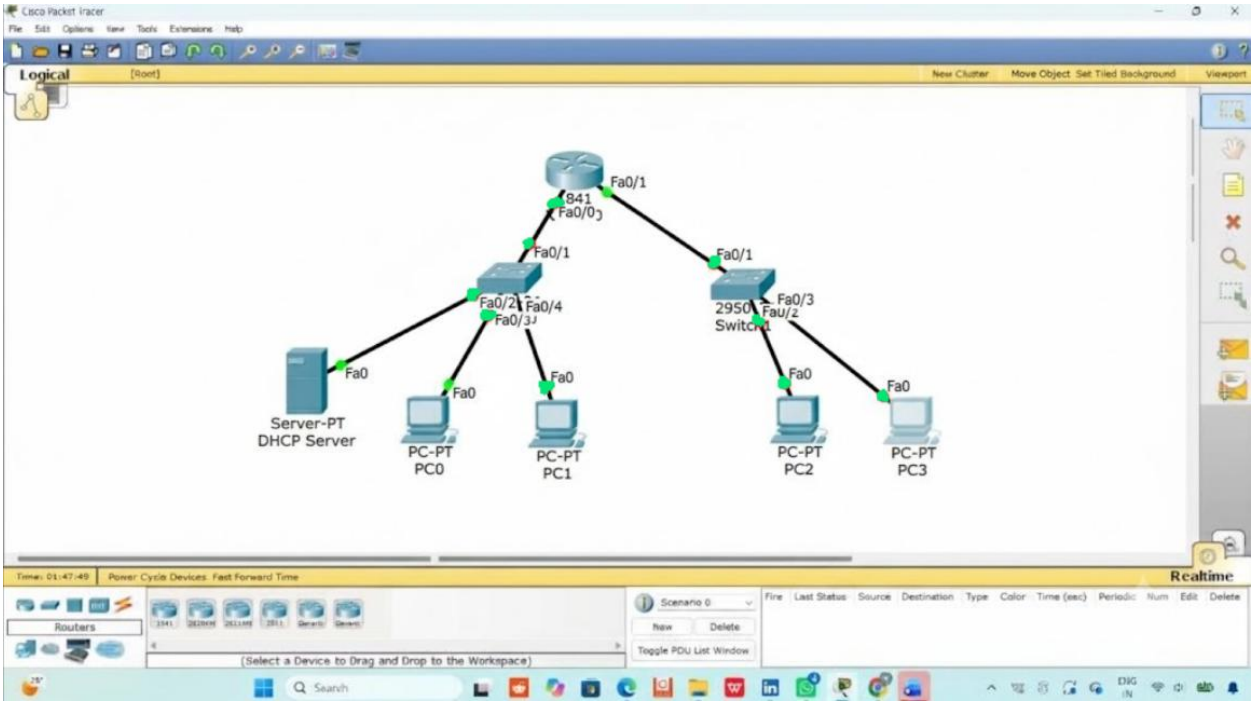
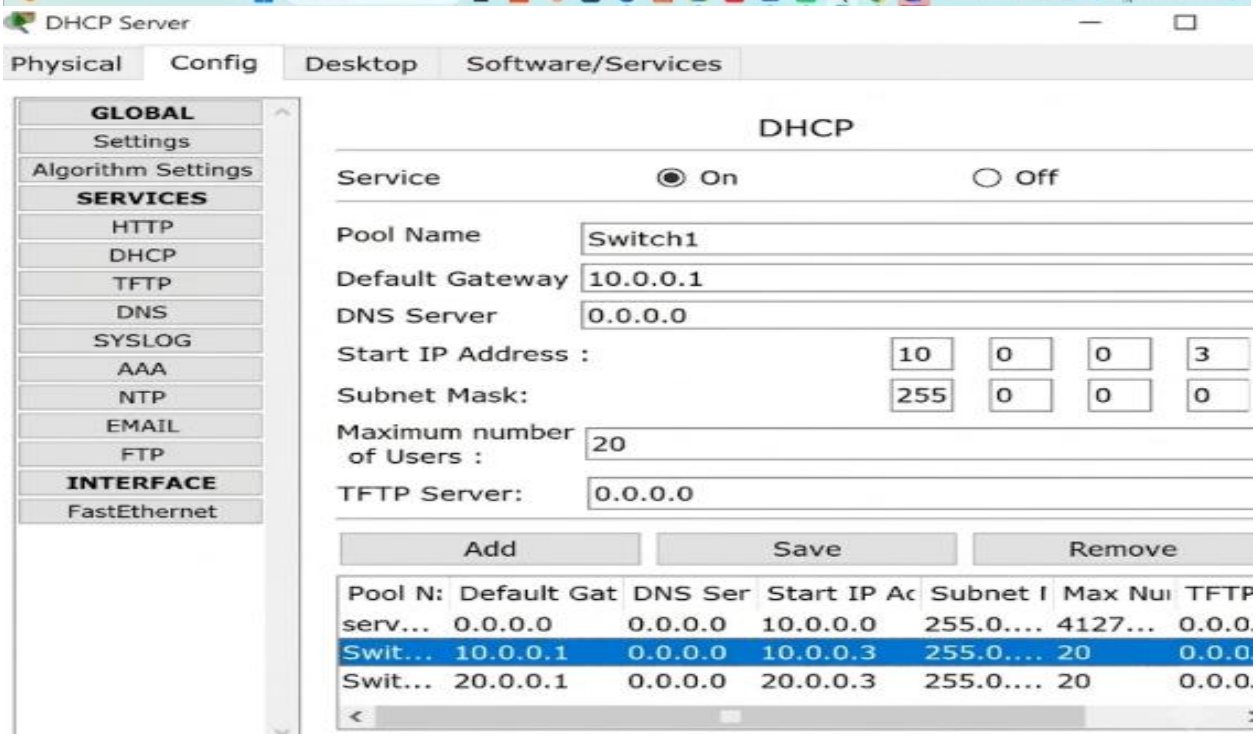
USN:1WN24CS186

3)How to configure DHCP within a LAN and Outside a LAN in a packet Tracer.





Router0

Physical | Config | CLI

IOS

```
Router>enable
Router#config t
% Invalid input detected at '^' marker.
                ^
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#ip helper-address 10.0.0.2
Router(config-if)#no shut

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

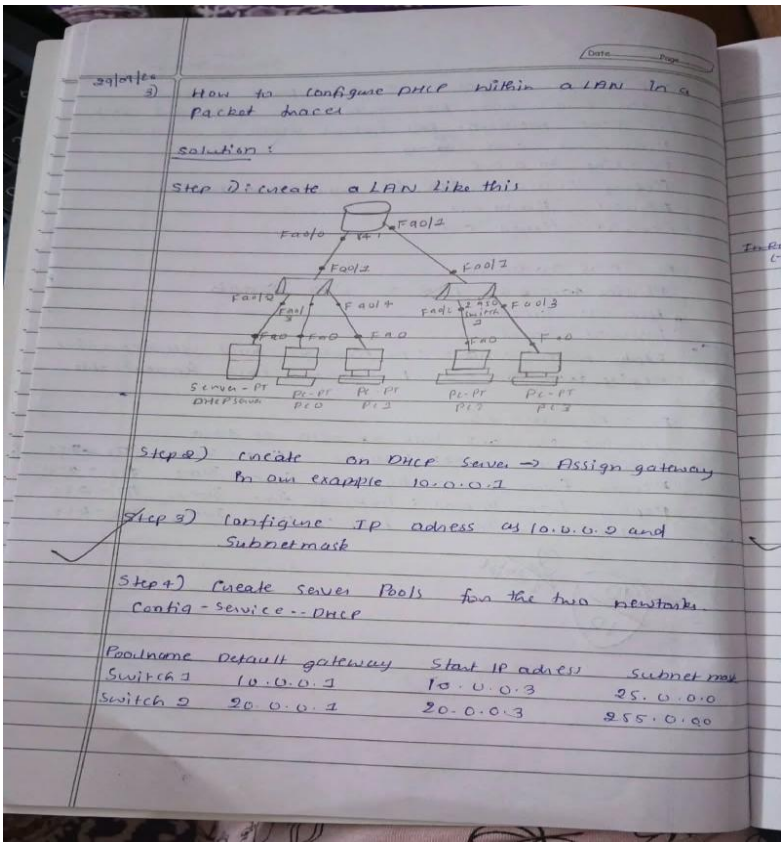
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#ip helper-address 10.0.0.2
Router(config-if)#no shut

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
o up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
```



Date Page

29/07/24

3) How to configure DHCP within a LAN in a Packet tracer

solution :

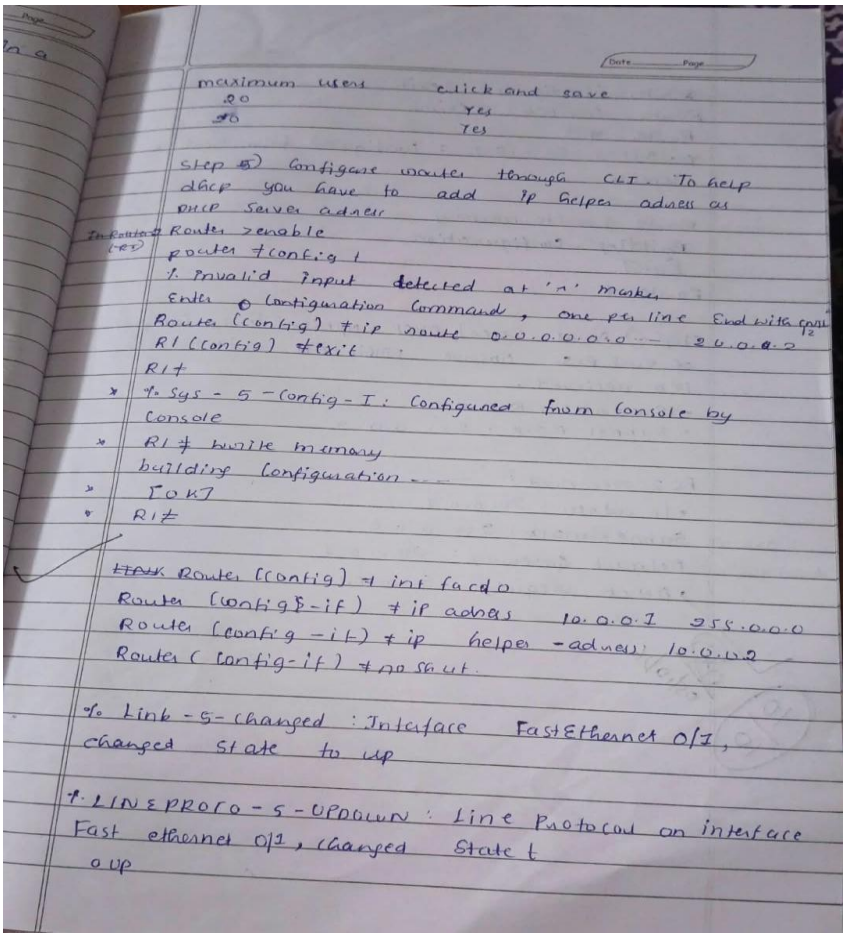
Step 1: create a LAN like this

Step 2) create an DHCP server → Assign gateway in our example 10.0.0.1

Step 3) Configure IP address as 10.0.0.2 and Subnetmask

Step 4) Create server Pools for the two networks
Config - service -- DHCP

Poolname	Default gateway	Start IP address	Subnet mask
Switch 1	10.0.0.1	10.0.0.3	25.0.0.0
Switch 2	20.0.0.1	20.0.0.3	255.0.00



maximum users — click and save
20 — Yes
20 — Yes

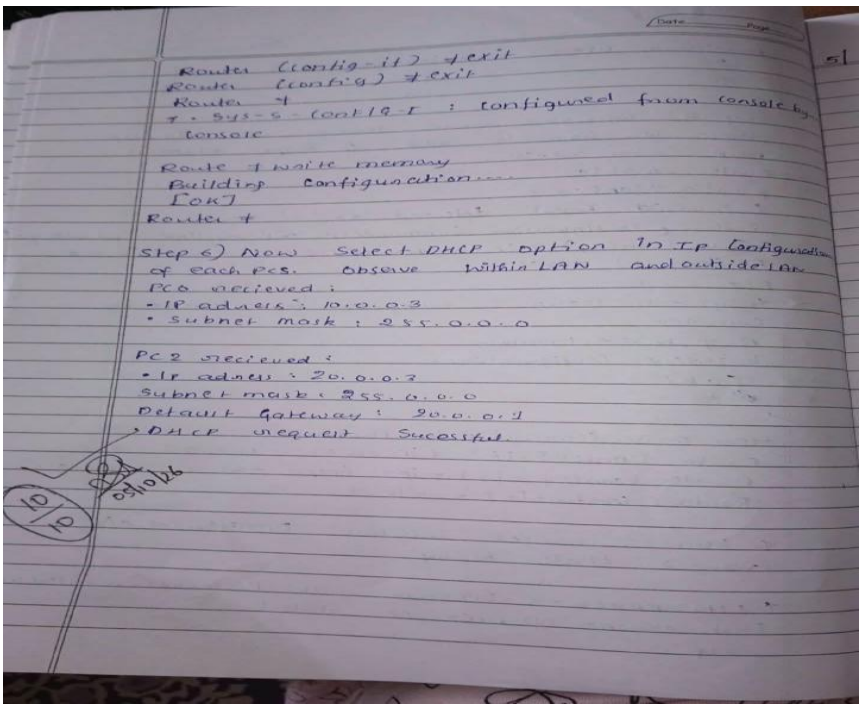
Step 5) Configure router through CLI. To help dhcp you have to add ip helper adress as DHCP Server adress

Router >enable
router #config t
% Invalid input detected at '^' marker.
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 — 20.0.0.2
R1(config)#exit
R1#
% Sys-5-config-I: Configured from console by console
R1# write memory
building configuration...
[OK]
R1#

Router(config)#int fa0/0
Router(config-if)#ip adress 10.0.0.1 255.0.0.0
Router(config-if)#ip helper-adress 10.0.0.2
Router(config-if)#no shut

% Link-5-changed: Interface FastEthernet 0/1, changed state to up

% LINEPROTO-5-UPDOWN: Line Protocol on interface Fast ethernet 0/1, changed state to up



Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: configured from console by console

Router#write memory
Building configuration...
[OK]
Router#

Step 6) Now select DHCP option in IP configuration of each PCs. Observe within LAN and outside LAN.

PC0 recieved:
- IP adress: 10.0.0.3
- Subnet mask: 255.0.0.0

PC2 recieved:
- Ip adress: 20.0.0.3
Subnet mask: 255.0.0.0
Default Gateway: 20.0.0.1
DHCP request Successful

10/10